GAUNTLET AI · TRAIN YOUR OWN SLM · RESULTS SUMMARY

Fine-tuning a 4B model for 10-digit US tariff classification, compared with a frontier model

Innocent Nortey · 2 November 2026 · Model: Qwen3-4B-Instruct-2507 + LoRA · Frontier reference: Claude Opus 5.5

In short: fine-tuning a 4B model on CBP rulings lifts its exact 10-digit accuracy from 42% to 60% with the same retrieved evidence, answers about 11× faster than a frontier model (0.9 s against 9.9 s per request) and costs about 1/150 as much per classification in bulk. Claude Opus 5.5 is still about 12 points more accurate, mostly because it remembers rare codes a small model has not seen.

42.0%	59.8%	71.6%	~11× · ~150×
Base Qwen3-4B, with retrieval	Fine-tuned Qwen3-4B, with retrieval	Claude Opus 5.5, with retrieval	Faster per request (H100) · cheaper per classification (batched) than Opus

Exact 10-digit accuracy on the final test: 500 CBP rulings issued Aug 2025 – Oct 2026, after all training data.

The question and setup

- **Question.** Does fine-tuning a small open model improve how well it assigns the exact 10-digit US tariff (HTSUS) code to a product, compared with the same base model and with a frontier model?
- **Open-set task.** The model chooses among about 19,800 tariff lines. Nothing in the input contains the answer. An earlier version of this project always placed the correct code among eight supplied options; that measured choosing, not classifying, so it was replaced.
- **Fresh, faithful test items.** 500 real CBP rulings issued after every ruling used in training, described in CBP's own words with codes, headings and the conclusion removed.
- **Fair comparison.** Every model receives identical input. The protocol and success criteria were frozen before any fine-tuned model was scored, and each arm was evaluated on the test set once.

Hypothesis supported

Fine-tuned – base: +17.8 points (95% CI +13.6 to +22.0). It also beats retrieval alone by +21.8 points (CI +17.6 to +26.0).

Frontier not matched

Fine-tuned – Opus 5.5: –11.8 points (95% CI –16.0 to –7.6), after three training attempts.

1 Retrieval: the search step

- **What it does.** Before any model answers, a hybrid search (keyword plus embeddings) returns 5 similar past CBP rulings with their codes and 20 candidate tariff lines. Every model sees exactly the same results.
- **How good it is.** The correct code is among the 20 candidates for 80% of test items. For the other 20%, a model can only be right if it knows the code itself.
- **Search alone,** taking its top candidate, is right 38.0% of the time.

2 Judgment: picking the right code from the evidence

Final test, same retrieved evidence	Exact 10-digit	When the right code was a candidate
Search alone (top candidate)	38.0%	—
Base Qwen3-4B	42.0%	52.5%
Fine-tuned Qwen3-4B	**59.8%**	**74.5%**
Claude Opus 5.5	71.6%	82.0%

Fine-tuning's main gain is here: when the right code is listed, the tuned model picks it 74.5% of the time against 52.5% for its base. Opus still chooses better by 7.5 points.

3 Memorization: knowing codes without search

Closed-book (no search results), ATLAS benchmark, 200 items	Exact 10-digit	6-digit
Base and fine-tuned Qwen3-4B, before memory training	0.0%	1–3%
Fine-tuned Qwen3-4B, after memory training	**12.5%**	**29.0%**
Claude Opus 5.5	41.0%	67.0%
ATLAS paper: fine-tuned LLaMA-3.3-70B	40.0%	57.5%
ATLAS paper: GPT-5-Thinking	25.0%	55.5%

- **Memory is where Opus wins most.** When search misses the right code, Opus still answers 30% of those items from memory; the fine-tuned model about 1%. That accounts for most of the 11.8-point overall gap.
- **Memory drills helped recall, not the main score.** Drilling every tariff line and ruling took the 4B model from 0% to 12.5% on ATLAS and to 45.7% closed-book on our own dev set. With search results present, they added only +0.7 points, because the remaining misses involve rare codes the model saw once or never (the typical code appears in only two training rulings).

4 Reasoning

- Opus reasons before answering; our model answers directly.
- We tried to teach reasoning with 488 Opus-written explanations. Each label came from CBP's own ruling, and each explanation was checked by a separate Codex judge that rejected all 50 deliberately broken test cases.
- The model learned the format but not good reasoning. Forcing it to reason first lowered accuracy by about 12 points, so the final model answers directly.

5 Missing information

- Opus flags a missing deciding fact on 38% of items, and the flag is informative: 60% accuracy on flagged items against 78% on the rest.
- Our model never flags, because no training example taught it to.

6 Cost and speed

One request at a time, warm GPU	Median wait	p90	Cost per request
Fine-tuned Qwen3-4B on an H100 (search included)	**0.89 s**	**0.99 s**	**≈ $0.001**
Fine-tuned Qwen3-4B on an A10G, the cheaper demo GPU	**1.74 s**	**1.89 s**	**≈ $0.0005**
Claude Opus 5.5 (API)	9.9 s	16.2 s	$0.034

- **About 11× faster on an H100, about 6× on an A10G.** Measured on 100 test items, one request at a time; the answers matched the batched evaluation exactly (66/100).
- **About 150× cheaper in bulk.** Processing 16 items at a time on an H100, each classification costs about $0.0002 of GPU time (0.2 s per item of throughput). Single requests cost more, about 35–65× less than Opus.
- Costs are GPU time at Modal list price; they exclude idle capacity and hosting.

Bottom line

Hypothesis supported. Fine-tuning a 4B model on CBP rulings substantially improves its classification, from 42% to 60% with the same retrieved evidence. The gain comes mainly from better judgment between candidate codes, at far lower cost than a frontier model.

Not at frontier level. Opus 5.5 is about 12 points more accurate overall: mostly because it has memorized rare codes a small model has not seen, plus somewhat better judgment and reasoning.

> **Not claimed:** that the fine-tuned model matches or beats Opus on accuracy or on memorization. On identical items (ATLAS, closed-book), Opus recalls more than three times as many exact codes.

Data: CBP CROSS rulings (Hugging Face corpus plus 15,500 rulings fetched from CROSS), USITC HTSUS export of 2026-10-30, ATLAS test split (arXiv 2509.18400). Training: Qwen3-4B-Instruct-2507, bf16 LoRA r=16 on a Modal H100; three training tries plus one memory follow-up judged on development data only. Full protocol, data checksums and reproducible commands: `docs/evals/results-v2.md` and `docs/evals/benchmark-protocol.md`.